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Dr. Ly

APBI 314

Assignment 4 - The science behind piglet deaths

Paper summaries:

1. First lactation performance of sows in Maternity Ring housing is comparable to that in farrowing crates

Plush et al. (2026) compared piglet mortality in conventional farrowing crates (FC) and Maternity Rings (MR). All sows were giving birth for the first time. The FC pens were 1.45 × 2.62 m, with a 0.60 × 1.80 m stall inside, so the sows could not turn around. The MR pens were 1.80 × 2.40 m and had an oval ring 0.25 m above the floor. This allowed the sows to turn around and helped protect the piglets. Among piglets born alive, mortality before weaning was similar in the two systems: 13.5% in FC and 13.8% in MR. However, more piglets died from crushing in MR, with an average of 0.9 piglets per litter compared with 0.3 in FC ($p < 0.001$). Before fostering, deaths also tended to be higher in MR, at 1.1 piglets per litter compared with 0.8 in FC, but this difference was not statistically significant ($p = 0.065$). Overall, MR allowed the sows more movement and had similar total piglet mortality, although more piglets died from crushing.

2. Impact of farrowing system and prepartum nest-building material on nest-building behaviour and farrowing in sows

Hukkinen et al. (2024) compared piglet mortality in two housing systems: free farrowing (FF) and temporary crating (TC). The study also provided straw, newspaper, or jute fabric as nesting materials. Both systems used similar pens, with 5.4 m² of space for the sow when the crate was open. In the FF system, sows could move freely during farrowing and

lactation, except when they needed to be briefly confined for handling. In the TC system, sows were kept in crates from two days before the expected farrowing date until three days after giving birth. After that, the crates were left open. When closed, the crates were 1.70–2.35 m long and 0.60–0.70 m wide, which limited the sow's movement. The results showed that TC sows were more likely to have at least one stillborn piglet in a litter compared with FF sows (odds ratio = 3.62, $p = 0.04$). However, TC sows were less likely to have two or more piglets crushed in a litter (odds ratio = 0.15, $p < 0.01$). There was no significant difference between the two systems in the chance of having at least two piglets die during the first three days after birth ($p = 0.19$). Overall, the study showed that temporary crating may help reduce piglet crushing, but it may also increase the risk of stillbirths.

3. Pen Versus Crate: A Comparative Study on the Effects of Different Farrowing Systems on Farrowing Performance, Colostrum Yield and Piglet Preweaning Mortality in Sows under Tropical Conditions

Dumniem et al. (2023) compared piglet mortality between conventional farrowing crates and free-farrowing pens. Both systems used pens with a total area of 4.7 m². In the crate system, sows were kept in a 1.80 × 0.60 m stall throughout the study. This gave them only 1.08 m² of space, so they could not turn around. In the free-farrowing system, the internal barrier was kept open, giving sows 3.25 m² of space and allowing them to turn around during farrowing and lactation. The results showed that piglet mortality before weaning was higher in free-farrowing pens than in crates (26.8% vs. 17.0%, $p = 0.045$). More piglets were also crushed by sows in free-farrowing pens (13.1% vs. 5.8%, $p = 0.037$). The sow's body condition also seemed to affect piglet survival. For sows with higher backfat thickness before giving birth, piglet mortality during the first 21 days was much higher in free-farrowing pens than in crates (37.8% vs. 10.9%, $p = 0.006$). Overall, the study suggests

that free-farrowing systems may lead to higher piglet mortality, especially for sows with more backfat before farrowing.

4. Impact of Duration of Farrowing Crate Closure on Physical Indicators of Sow Welfare and Piglet Mortality

Ceballos et al. (2021) studied how opening the crate at different times affected piglet mortality. All sows were kept in pens measuring 2.36 × 1.70 m, with a total area of about 4.0 m². Each pen had an internal crate measuring 0.635 × 2.13 m. The sows were divided into three groups: those kept in crates until weaning (PC), those released on day 4 after farrowing (T4), and those released on day 7 (T7). Once the crates were opened, the sows could turn around freely for the rest of lactation. The results showed that total piglet mortality was higher in T4 (27.8%) than in PC (25.9%, $p = 0.03$) and T7 (23.9%, $p = 0.04$). However, there was no significant difference between T7 and PC ($p = 0.94$). From birth to weaning, piglets in T4 also had a higher rate of death from crushing compared with PC (incidence rate ratio = 1.30, $p = 0.04$). There was no significant difference in crushing deaths between T7 and PC (incidence rate ratio = 1.17, $p = 0.54$). Overall, the study suggests that opening the crate on day 7 may allow sows more freedom to move without significantly increasing piglet mortality compared with keeping them in crates until weaning.

Conclusion:

So, these studies show that the effects of farrowing housing on piglet mortality depend on the housing system and how it is managed.Giving sows more space to move can increase the risk of piglets being crushed (Plush et al., 2026; Dumniem et al., 2023). However, the results for overall piglet mortality were different. Plush et al. (2026) found similar mortality in both systems, while Dumniem et al. (2023) found higher mortality in

free-farrowing pens. Different causes of piglet death are also important to consider. Hukkinen et al. (2024) found that temporary crating reduced the chance of having multiple piglets crushed in a litter, but increased the chance of stillborn piglets. The timing of opening the crates also made a difference. Ceballos et al. (2021) found that opening crates on day 7 resulted in similar total piglet mortality to keeping them closed until weaning, while opening them on day 4 led to higher mortality. The sow's body condition may also affect the results, since sows with more backfat had higher piglet mortality in free-farrowing pens than in crates (Dumniem et al., 2023). Differences in pen design, sow characteristics, and how piglet mortality was measured also make it difficult to directly compare these studies. Overall, some housing systems that give sows more freedom can have similar piglet survival rates to conventional crates, but reducing piglet crushing and considering the sow's body condition are still important.

References

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